

Jump Process

(1. Stochastics with jumps)

Why jumps ?

Three main reasons why jumps improve capturing real-world dynamics

- I) **Black-Scholes** assumes stock price evolves continuously i.e.; price cannot jump suddenly. But real markets often exhibit **sudden jumps** which cannot be explained by Brownian shocks. Large price moves can occur in a short time due to defaults, crashes, geopolitical shocks, central bank policies etc....
- II) Stock returns often exhibit **fat tails** (leptokurtic) compared to the normal distribution assumption as enforced by Brownian motion. Jumps can increase the probability of extreme outcomes.
- III) Jumps help explain volatility smile/skew.

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \quad \longrightarrow \quad \frac{dS_t}{S_t} = \mu dt + \sigma dW_t + \underbrace{(Y - 1)dN_t}_{\text{(jump)}}$$



Recap Poisson Process

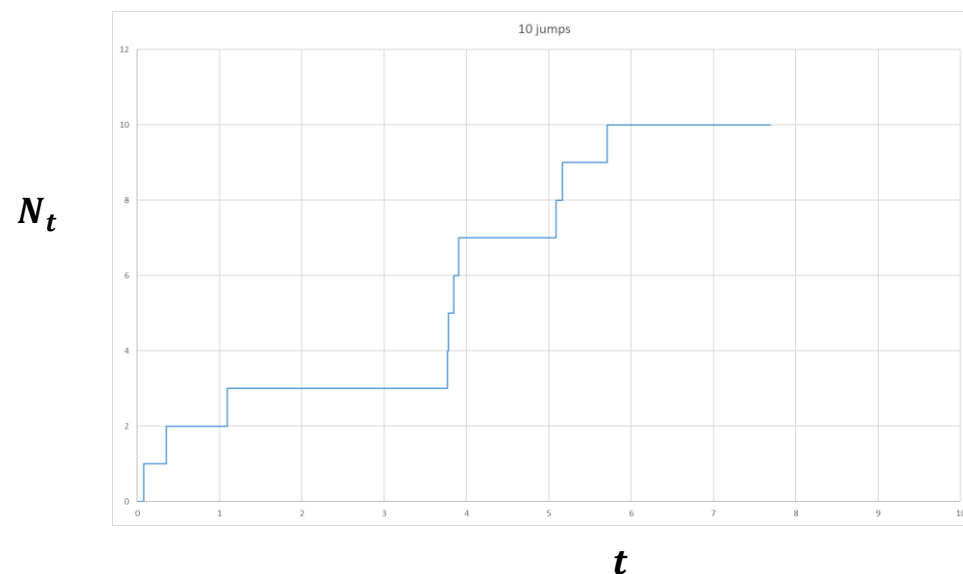


The most elementary jump process is a **Poisson process** N_t which is a **counting process** with the following properties

1. $N_0 = 0$

2. $\mathbb{P}(N_t - N_s = k) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}$ (increments $N_t - N_s$ are independent and follow Poisson distribution with parameter $\lambda(t-s)$)

Each arrival (jump) is of size 1. The time between two arrivals follows exponential distribution with parameter λ



Compensated Poisson Process

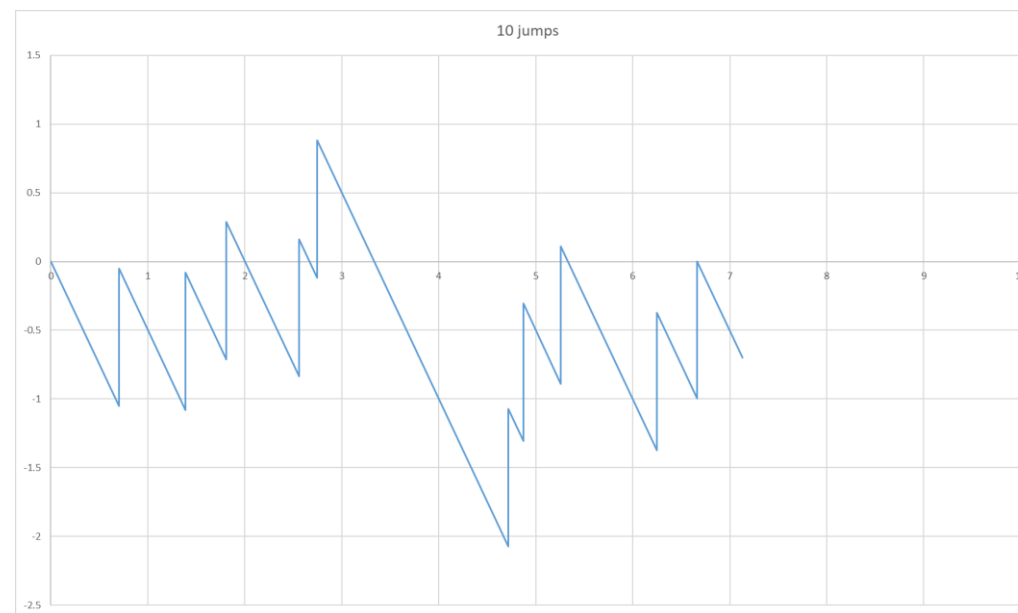
What is the mean and variance of the counting process N_t ?

$$\mathbb{E}[N_t] = \lambda t$$

$$\mathbb{V}[N_t] = \lambda t$$

The counting process has a drift of λt . We define the **Compensated Poisson** process as $M_t = N_t - \lambda t$

Note that M_t is a martingale.



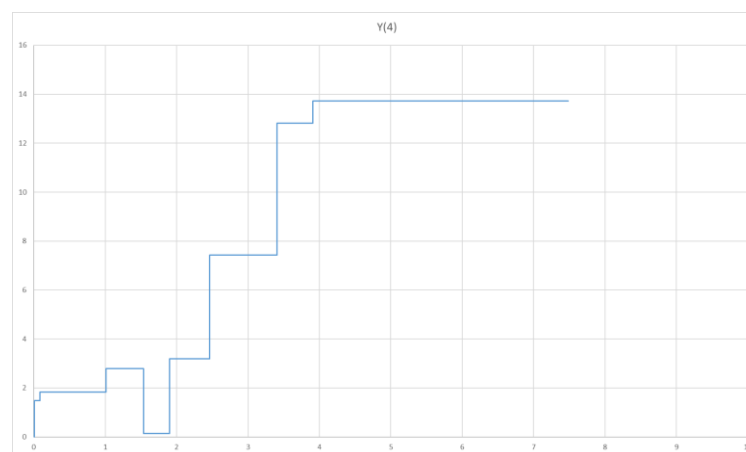
Compound Poisson Process



A plain counting process does not give much flexibility as the jump size = 1. We would be better off if we could model the size of each jump as well. Let's consider a sequence of $\{Z\}_i$ as a sequence of **i.i.d** random variables which denote the size of the jump. The **Compound Poisson** process is defined as the sum of a random number of such jumps.

$$Y_t = Z_1 + Z_2 + \dots + Z_{N(t)} = \sum_{i=1}^{N(t)} Z_i$$

- 1) Show that the mean of Y_t is $\mathbb{E}[Y_t] = \lambda t \mathbb{E}[Z]$ (Hint : Use Tower property of iterative expectation)
- 2) What would be the expression for compensated compound Poisson process ?



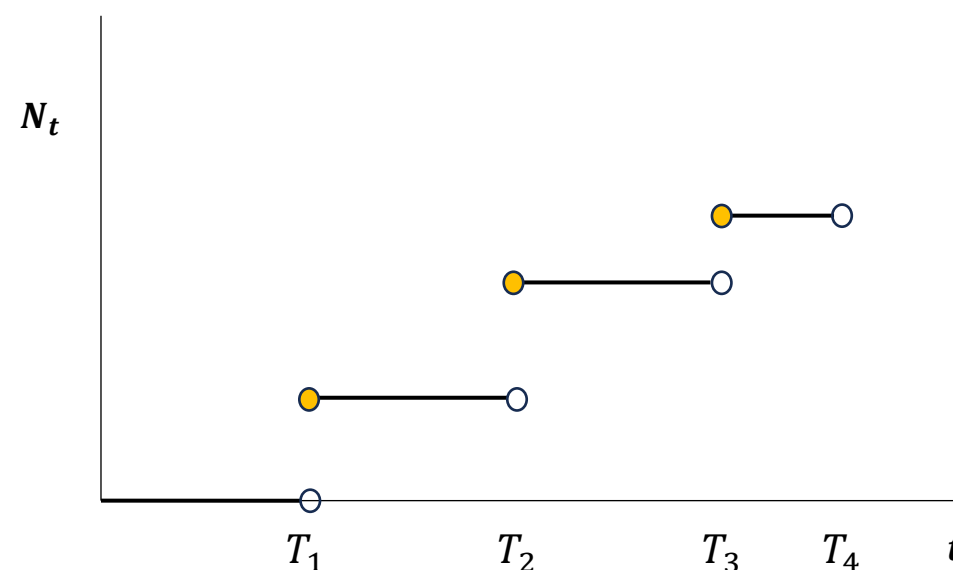
Cadlag (RCLL)



In mathematics, a **cadlag** or **RCLL** (right-continuous with left limits) is a function defined in $\mathbb{R}$ which is right-continuous everywhere and left limit exists everywhere.

left limit exists $f(t^-) := \lim_{s \rightarrow t^-} f(s)$ exists

right limits exists $f(t^+) := \lim_{s \rightarrow t^+} f(s)$ exists and is equal to $f(t)$



Note : the jump gets added to the t^- instant as t already includes the jump

$$N_t = N_{t^-} + Z(t)$$

Infinitesimal time behavior

How does the distribution look like in infinitesimal time (dt) ?

$$\mathbb{P}(N_{t+dt} - N_t = 1) = e^{-\lambda dt} \frac{(\lambda dt)^1}{1!} = \lambda dt + o(dt)$$

$$\mathbb{P}(N_{t+dt} - N_t = 0) = e^{-\lambda dt} \frac{(\lambda dt)^0}{0!} = 1 - \lambda dt + o(dt)$$

$$\mathbb{P}(N_{t+dt} - N_t > 1) = o(dt)$$

Essentially the Poisson increment in infinitesimal period behaves like a **Bernoulli random**

$$dN_t = \begin{cases} 1, & \text{prob} = \lambda dt \\ 0, & \text{prob} = 1 - \lambda dt \end{cases}$$

Ito's Lemma for Poisson Process – (I)

Just like we performed Ito calculus on Brownian motion and functions on Brownian motion, we need to develop the machinery for jump process. Consider $N(t)$ as our counting process with intensity λ . How do we develop the SDE for a function that depends on $N(t)$?

$$t \rightarrow t + dt \quad dN_t = \begin{cases} 1, & \text{prob} = \lambda dt \\ 0, & \text{prob} = 1 - \lambda dt \end{cases} \quad df(N_t) = ?$$

$$\begin{aligned} df(N_t) &= f(N_{t+dt}) - f(N_t) \\ &= f(N_t + dN_t) - f(N_t) \\ &= [f(N_t + 1)dN_t + f(N_t + 0)(1 - dN_t)] - f(N_t) \\ &= [f(N_t + 1) - f(N_t)]dN_t \end{aligned}$$

$$df(N_t) = [f(N_t + 1) - f(N_t)]dN_t$$

Note that it's a full revaluation as the jump is of finite size (not infinitesimal)



Ito's Lemma for Poisson Process – (II)

Now consider a function that depends on both t and N_t

$$t \rightarrow t + dt \quad dN_t = \begin{cases} 1, & \text{prob} = \lambda dt \\ 0, & \text{prob} = 1 - \lambda dt \end{cases} \quad df(t, N_t) = ?$$

$$df(t, N_t) = \frac{\partial f(t, N_t)}{\partial t} dt + [f(t, N_t + 1) - f(t, N_t)] dN_t$$

The change comes from two sources; explicit time dependence (continuous drift) + probabilistic 1 jump



Ito's Lemma for Poisson Process – (III)

Now consider a more general stochastic process X_t and a function $f(X_t)$ with dynamics

$$dX_t = \mu(t, X_t)dt + v(t, X_t)dN_t$$

$$df(t, X_t) = \left[\frac{\partial f(t, X_t)}{\partial t} + \frac{\partial f(t, X_t)}{\partial X_t} \mu(t, X_t) \right] dt + [f(t, X_{t-} + v(t, X_t)) - f(t, X_t)] dN_t$$

It's very important to note the use of time instant in the above equation. The jump always needs to be added to X_{t-} as X_t already contains the jump (RCLL).



Ito's Lemma for Jump Diffusion models – (IV)

We can also add a diffusion term

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t + v(t, X_t)dN_t$$

$$df(t, X_t) = \underbrace{\left[\frac{\partial f(t, X_t)}{\partial t} + \frac{\partial f(t, X_t)}{\partial X_t} \mu(t, X_t) + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 f(t, X_t)}{\partial x^2} \right] dt + \sigma(t, X_t) \frac{\partial f(t, X_t)}{\partial X_t} dW_t}_{\text{(continuous part)}} + \underbrace{[f(t, X_{t-} + v(t, X_t)) - f(t, X_t)] dN_t}_{\text{(discrete jump impact)}}$$



Ito's table

Now that we have a new source of randomness, we want to enhance the Ito table for product rule.

	dt	dW	dN
dt	0	0	0
dW	0	dt	0
dN	0	0	dN

1) Prove $dN^2 = dN$

2) If $dX_t = adt + bdW_t + cdN_t$ & $dY_t = edt + fdW_t + gdN_t$, then what is $dX_t \cdot dY_t = ?$



Exercises



1. Two assets can jump with intensities λ_1 and λ_2 , respectively.
 - a) What is the probability that asset 1 jumps first ?
 - b) What is the probability that we observe 2 jumps from asset 1 before 2 jumps from asset 2 ?

